

LEVERAGING UNSUPERVISED LEARNING TO DRIVE BUSINESS GROWTH



STRATEGIC CUSTOMER SEGMENTATION FOR MARKETING OPTIMIZATION

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PROBLEM DEFINITION



The Challenge

Inefficient outreach through mass marketing.
High operational costs due to lack of targeting.

Business Impact

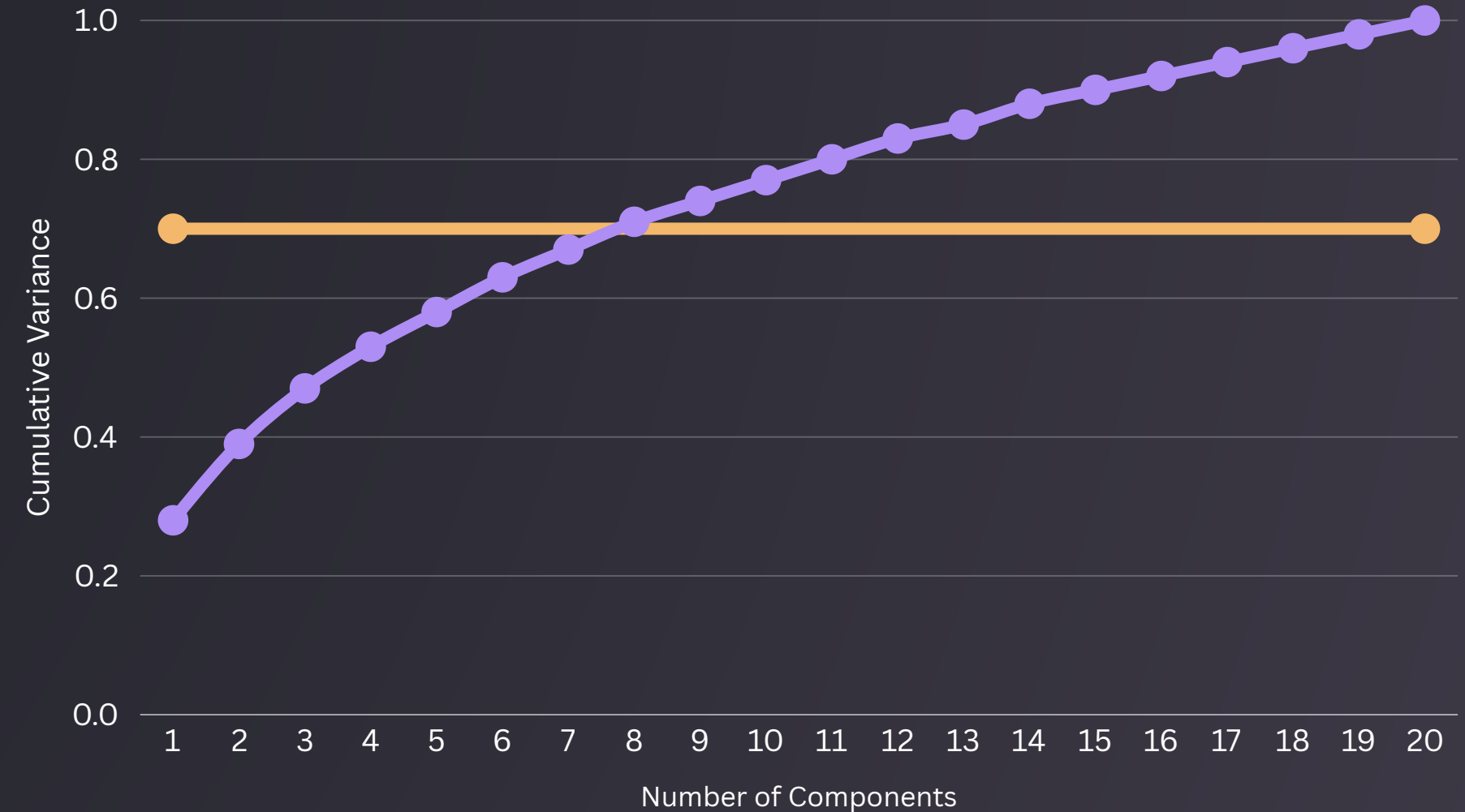
Budget waste on non-converting leads and rising customer engagement fatigue.

Objective

Transition from generic campaigns to **data-driven, hyper-personalized segments** to maximize **ROI**.

DATA & METHODOLOGY

PCA - Cumulative Explained Variance



Dataset Overview

2,240 customers with **29** behavioral and demographic features.

Key Pre-processing

Handling missing values (Income) and removing demographic outliers.
Feature engineering: Calculating "**Total Spending**" and "**Household Size**".

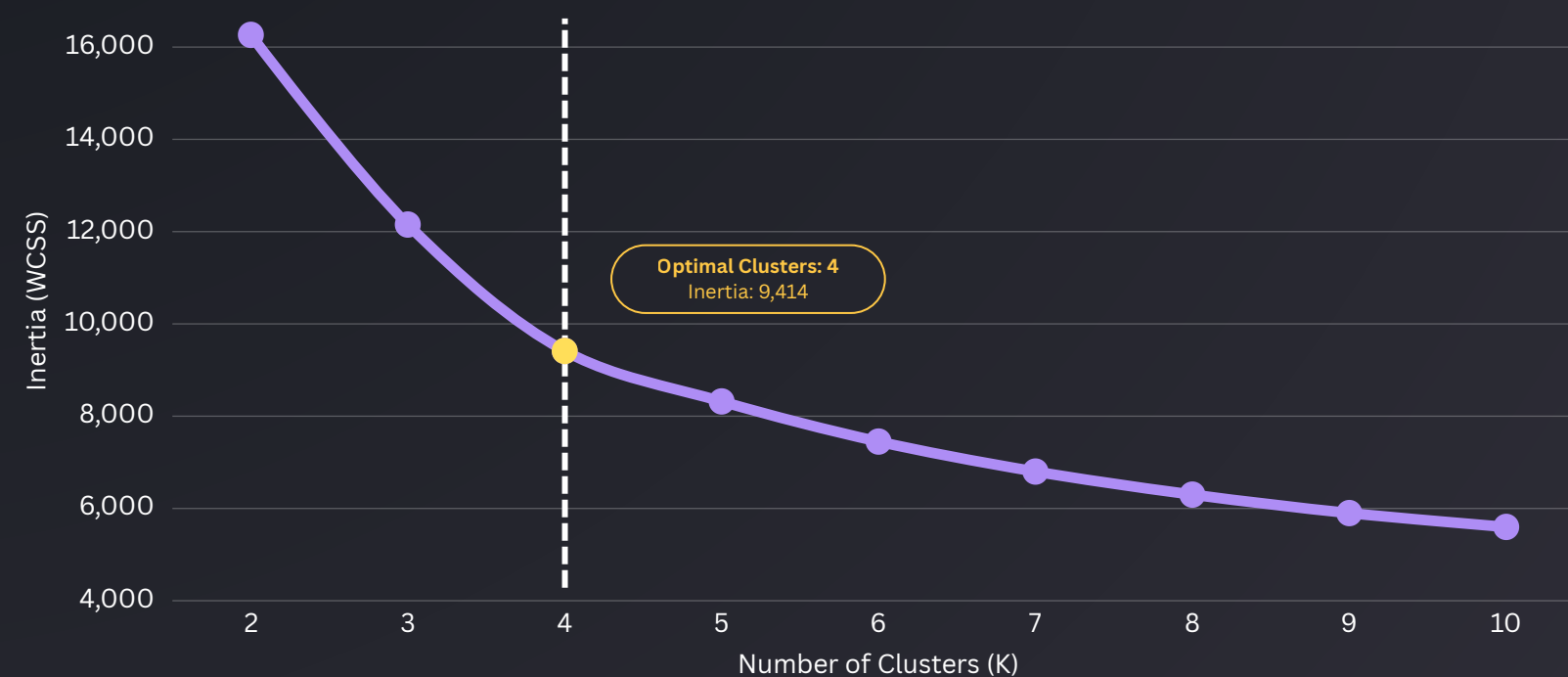
Dimensionality Reduction

Using **PCA** (Principal Component Analysis) to eliminate noise and focus on the core drivers of variance.

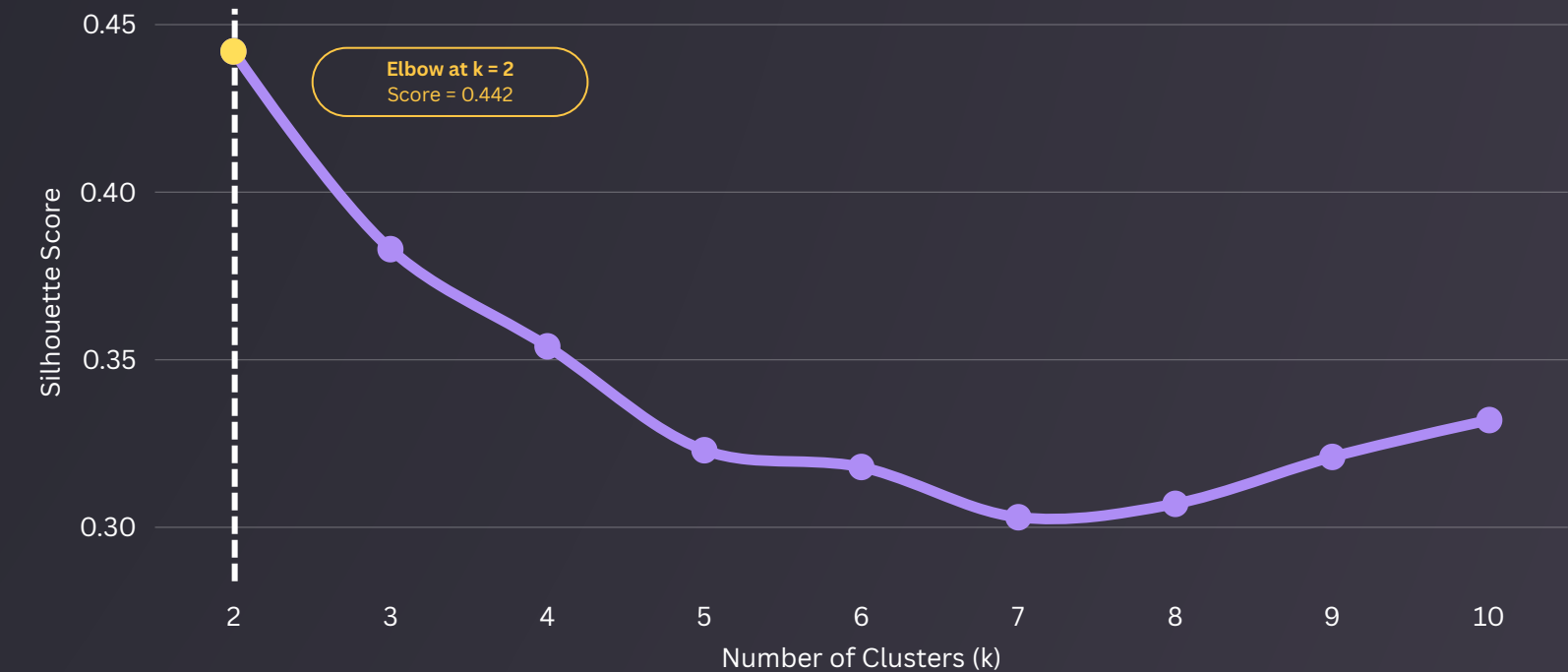
SOLUTION DESIGN

THE MODELS

Elbow Method for Optimal K



Silhouette Score Analysis for KMeans Clustering



Algorithm Comparison

Tested K-Means, K-Medoids, Agglomerative Clustering, and GMM.

Selection Criteria

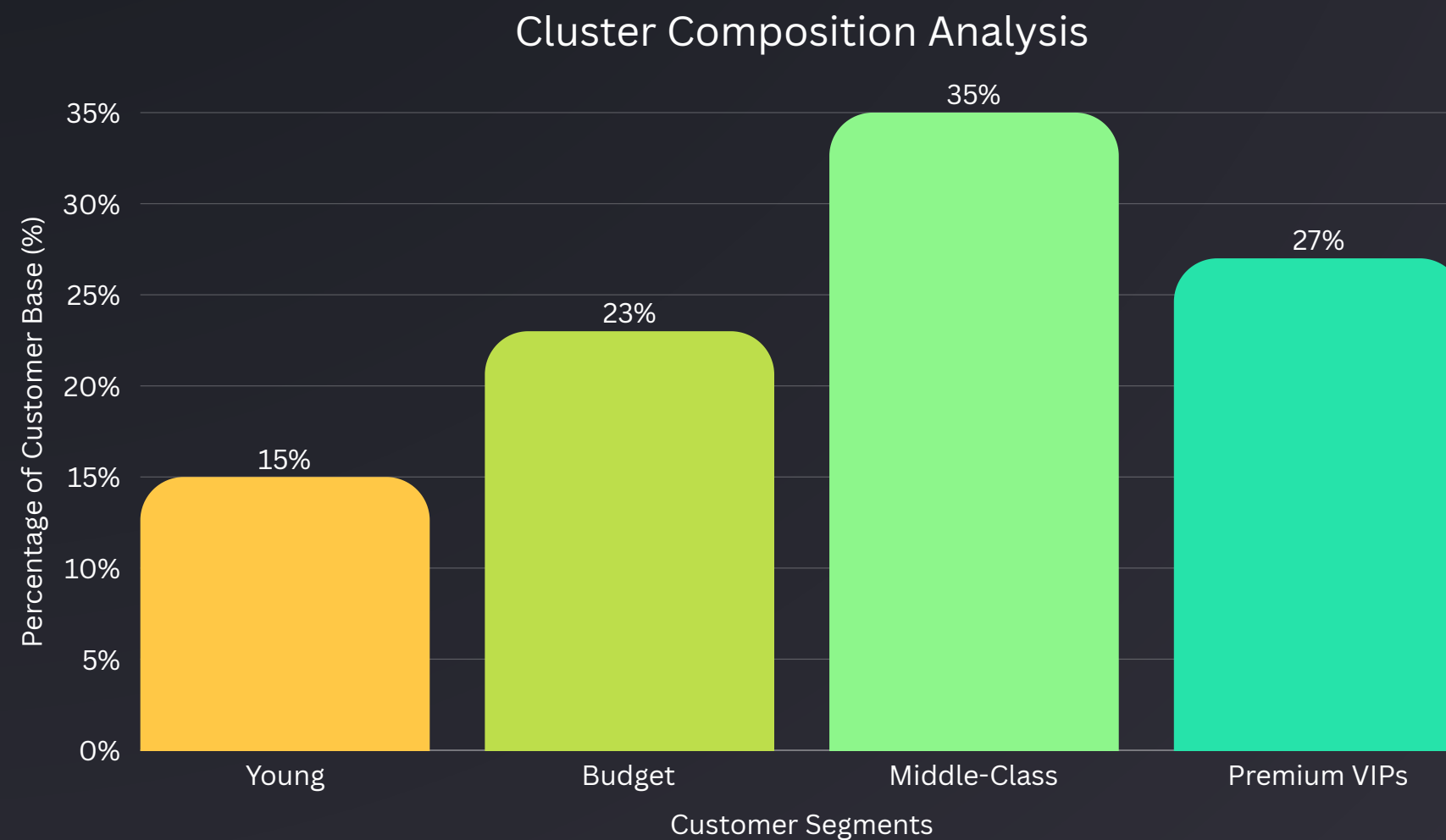
Elbow Method and Silhouette Coefficient.

Final Decision

K-Means (K=4) was chosen for its high stability and clear business interpretability.

THE 4 PERSONAS

KEY FINDINGS



N=2,240 customers. Segments derived from K-Means (K=4) clustering.

Avg. Income: \$78k | Avg. Spend: \$1.2k

Premium VIPs

High income, high spending in luxury categories like Wine/Meat.



Avg. Income: \$52k
High Purchase Frequency

Middle-Class Mainstay

Reliable frequent buyers, focus on balanced spending.



Avg. Family Size: 4.2 | High Deal Usage

Budget-Conscious Families

Large households, price-sensitive, focus on essentials.



Avg. Age: 28 | High Web Interaction

Young Starters

Low current spending but high digital engagement and future growth potential.



BUSINESS RECOMMENDATIONS

Premium VIPs

Retention & Exclusivity

Implement exclusive loyalty programs, early access to new collections, and concierge services to maximize Lifetime Value (LTV).



Middle-Class Mainstay

Value & Bundling

Focus on "Family Value Packs" and household essentials. Use volume-based discounts to increase the average basket size.



Primary Goal

To transform customer insights into a competitive advantage by delivering hyper-personalized experiences that increase **ROI** and minimize marketing waste.

Budget-Conscious Families

Re-engagement

Use personalized discount triggers and "we miss you" email automation based on their last purchase date to prevent churn.



Young Starters

Digital Engagement

Deploy mobile-first social media campaigns and influencer partnerships to build long-term brand affinity with this tech-savvy group.



COST-BENEFIT & RISKS



Projected Benefits

Estimated **15-20%** increase in campaign **ROI** based on improved targeting.



Implementation Costs

Low technical overhead (**CRM integration** and marketing creative updates).



Risks

"**Cluster Drift**" due to economic changes; requires model retraining every **6 months**.



QUESTIONS?

THANK YOU
FOR YOUR ATTENTION



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